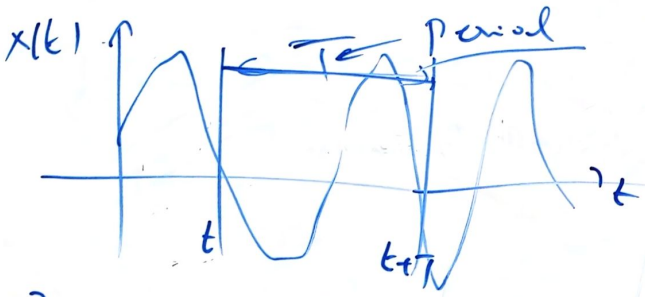
23/6/26

# L2 — Waves — Simple Harmonic Motion/Oscillation (SHM/SHO)




Periodic motion: $x(t+T) = x(t)$

Simplest periodic motion:

$x(t) = A \sin \omega t$

$x(t+T) = x(t) =$

$A \sin(\omega(t+T)) = A \sin(\omega t)$ $(A \neq 0)$

$\sin(\theta + \alpha) = \sin\theta$

$\theta = \omega t, \quad \alpha = \omega T$

$\Rightarrow \alpha = 0, \pm 2\pi, \pm 4\pi \ldots$

$\alpha = 2\pi \equiv \omega T$

$T = \frac{2\pi}{\omega}$.

$\Rightarrow T$ (secs) the object goes thro' 1 cycle of motion / osc.

In 1 sec:

$\#\,osc = \frac{1}{T} = \nu$

frequency

$\nu = \frac{1}{T} = \frac{\omega}{2\pi}$

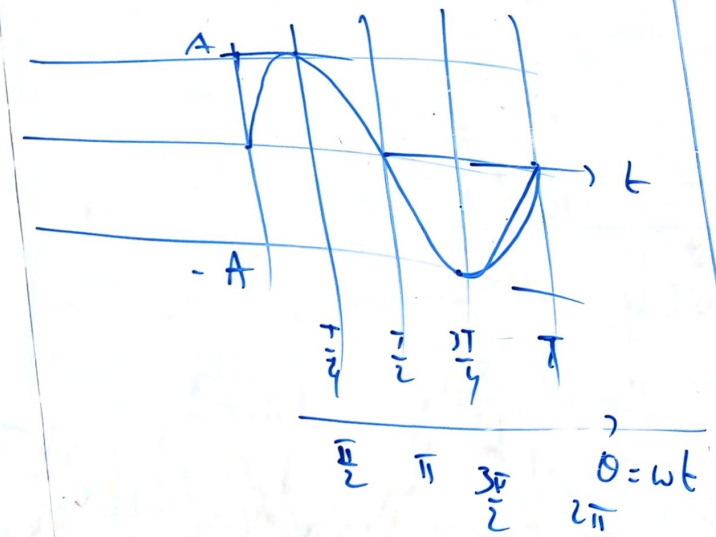
$\omega = 2\pi\nu$

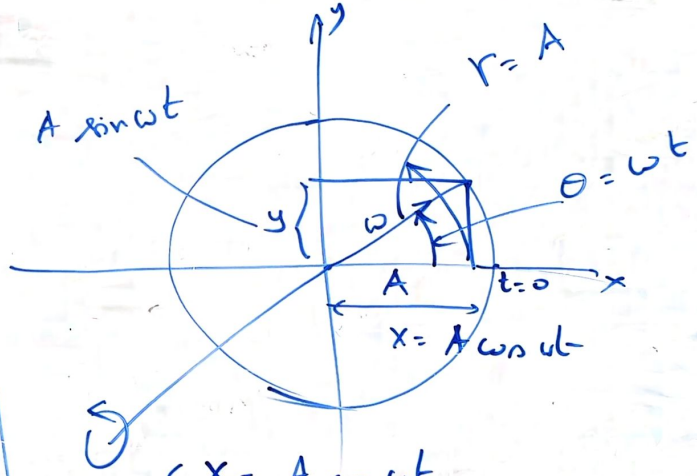
: angular frequency

$\max(\sin\omega t) = +1$

$\min(\sin\omega t) = -1$

$\begin{Bmatrix} \max \\ \min \end{Bmatrix} (A\sin\omega t) = \pm A$




$\begin{cases} x = A\cos\omega t \\ y = A\sin\omega t \end{cases}$

Starting pt.: $x = A,\ y = 0$ $(t=0)$

$x = A\cos(\omega t + \theta_0)$

$\boxed{y = A\sin(\omega t + \theta_0)}$

$r(t) = A$

$t = 0$: $\theta(0) = \theta_0 \neq 0$

$\Delta\theta = \theta - \theta_0 = \omega t$

$\boxed{\frac{d\theta}{dt} = \omega}$

$\int_{\theta_0}^{\theta} d\theta = \omega \int_0^t dt$

$\Delta\theta = \theta - \theta_0 = \omega t$

$\theta(t) = \theta_0 + \omega t$

$0 \leq \theta < 2\pi$

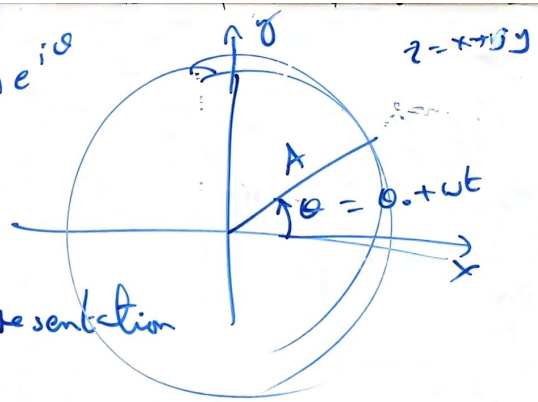
$\theta \in [0, 2\pi)$

31/8/20

# L2 — Waves — Simple Harmonic Motion / Oscillation (SHM / SHO)

$$\left.\begin{matrix} A\cos\omega t \\ A\sin\omega t \end{matrix}\right\} \longrightarrow \vec{r} = A\cos\omega t\,\hat{x} + A\sin\omega t\,\hat{y}$$

$\hat{x}$ ($\hat{i}$), $\hat{y}$ ($\hat{j}$) — Vector representation




$z = Ae^{i\theta}$

$$z = x + iy$$
$$= A\cos\omega t + iA\sin\omega t$$
$$= A(\cos\omega t + i\sin\omega t)$$
$$= Ae^{i\omega t} = Ae^{i\theta}$$

$e^{i\theta} = \cos\theta + i\sin\theta$

$\theta = \theta_0 + \omega t$

$z = x + iy = Ae^{i(\omega t + \theta_0)}$

$x = Re[z] = A\cos(\omega t + \theta_0)$

$y = Im[z] = A\sin(\omega t + \theta_0)$

→ A complex rotating vector or phasor representation.

$x = A \sin(\omega t + \varphi)$

$\frac{dx}{dt} = A\left(\frac{d}{d\theta}\sin\theta\right)\left(\frac{d(\omega t+\varphi)}{dt}\right)$ ($\omega$)

$= A\omega \cos(\omega t + \varphi)$

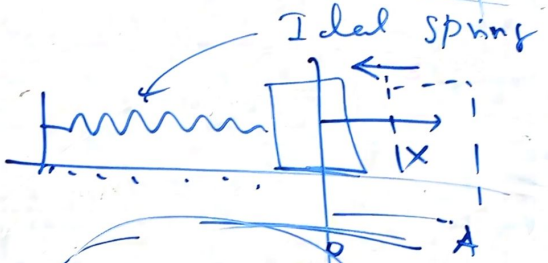
$\frac{d^2x}{dt^2} = -A\omega^2 \sin(\omega t + \varphi)$

$\boxed{\frac{d^2}{dt^2}x(t) = -\omega^2 x(t)}$

$a \propto -x$

$x = A\cos(\omega t + \theta_0)$

$\theta_0 = \varphi - \frac{\pi}{2}$




$F = -kx$

$m\frac{d^2x}{dt^2} = -kx$

$\therefore \frac{d^2x}{dt^2} = -\frac{k}{m}x \Rightarrow \boxed{\frac{k}{m} = \omega^2}$

$\omega = \sqrt{\frac{k}{m}}$

$\omega^2 = \frac{\text{Restoring force per unit displacement}}{m}$